Simplifying Surds

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS surd *(pūtake tauoti-kore)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Square numbers and square roots ●○○

1. Write the value of each square root.

a) $\sqrt{16}$ = _____ **b)** $\sqrt{49}$ = _____ **c)** $\sqrt{81}$ = _____ **d)** $\sqrt{121}$ = _____

e) $\sqrt{144}$ = _____ **f)** $\sqrt{225}$ = _____ **g)** $\sqrt{400}$ = _____ **h)** $\sqrt{900}$ = _____

2. Write the largest square number that divides into each number.

a) 12 _____ **b)** 18 _____ **c)** 50 _____ **d)** 48 _____

e) 72 _____ **f)** 75 _____ **g)** 98 _____ **h)** 200 _____

B Simplifying a surd ●○○

Remember: $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$. Look for the largest square number that divides in.

3. Simplify.

a) $\sqrt{12}$ = _____ **b)** $\sqrt{18}$ = _____ **c)** $\sqrt{20}$ = _____ **d)** $\sqrt{45}$ = _____

e) $\sqrt{50}$ = _____ **f)** $\sqrt{27}$ = _____ **g)** $\sqrt{72}$ = _____ **h)** $\sqrt{48}$ = _____

i) $\sqrt{98}$ = _____ **j)** $\sqrt{200}$ = _____ **k)** $\sqrt{125}$ = _____ **l)** $\sqrt{147}$ = _____

4. Simplify.

a) $2\sqrt{12}$ = _____ **b)** $3\sqrt{50}$ = _____ **c)** $5\sqrt{8}$ = _____ **d)** $4\sqrt{27}$ = _____

C Adding and subtracting surds ●●○

Watch out: Only like surds can be added or subtracted, so simplify each surd first.

5. Simplify.

a) $3\sqrt{2} + 5\sqrt{2}$ **b)** $7\sqrt{5} - 2\sqrt{5}$ **c)** $4\sqrt{3} + \sqrt{3}$

d) $\sqrt{8} + \sqrt{2}$ **e)** $\sqrt{12} + \sqrt{27}$ **f)** $\sqrt{50} - \sqrt{18}$

g) $\sqrt{45} + \sqrt{20}$ **h)** $2\sqrt{3} + \sqrt{75}$ **i)** $\sqrt{72} - \sqrt{32}$

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D Multiplying and dividing surds ●●○

Remember: $\sqrt{a} \times \sqrt{b} = \sqrt{a \times b}$ and $\sqrt{a} \div \sqrt{b} = \sqrt{a \div b}$

6. Simplify.

a) $\sqrt{2} \times \sqrt{8}$ **b)** $\sqrt{3} \times \sqrt{5}$ **c)** $\sqrt{6} \times \sqrt{3}$

d) $2\sqrt{3} \times 4\sqrt{3}$ **e)** $3\sqrt{2} \times 2\sqrt{5}$ **f)** $\sqrt{7} \times \sqrt{7}$

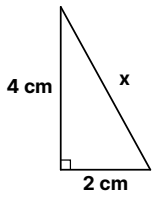
g) $\sqrt{20} \div \sqrt{5}$ **h)** $\sqrt{48} \div \sqrt{3}$ **i)** $6\sqrt{10} \div 2\sqrt{2}$

j) $2\sqrt{6} \times 3\sqrt{2}$ **k)** $\sqrt{72} \div \sqrt{2}$ **l)** $\sqrt{5} \times \sqrt{10}$

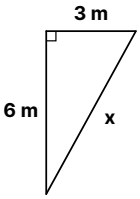
E Exact lengths ●●●

7. Find the length of the side marked x. Give each answer as a surd in simplest form. Diagrams are not to scale.

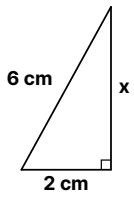
a)



b)



c)



F Mixed practice ●●●

8. Expand and simplify.

a) $\sqrt{2}(\sqrt{2} + 5)$ **b)** $\sqrt{3}(\sqrt{12} - 1)$ **c)** $(\sqrt{5} + 2)(\sqrt{5} - 2)$

d) $(1 + \sqrt{3})^2$ **e)** $2\sqrt{3}(\sqrt{3} + \sqrt{12})$ **f)** $(\sqrt{6} + \sqrt{2})(\sqrt{6} - \sqrt{2})$

9. A rectangle is $\sqrt{12}$ cm long and $\sqrt{3}$ cm wide. Give each answer in simplest form.

a) Find its area. **b)** Find its perimeter.

★ Extension

10. Rationalise the denominator: write each fraction without a surd on the bottom.

a) $\dfrac{1}{\sqrt{2}}$ **b)** $\dfrac{6}{\sqrt{3}}$ **c)** $\dfrac{10}{\sqrt{5}}$ **d)** $\dfrac{3}{2\sqrt{3}}$

11. A square has an area of 48 cm². Write the length of its sides in simplest surd form.

12. The two shorter sides of a right-angled triangle are $\sqrt{6}$ cm and $\sqrt{10}$ cm long. Find the length of the hypotenuse.

13. Show that $\sqrt{50} + \sqrt{8} = 7\sqrt{2}$.

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Simplifying Surds

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS surd *(pūtake tauoti-kore)* · square number *(tau pūrua)* · square root *(pūtakerua)*

A Square numbers and square roots ●○○

1. Write the value of each square root.

a) $\sqrt{25}$ = _____ **b)** $\sqrt{36}$ = _____ **c)** $\sqrt{64}$ = _____ **d)** $\sqrt{100}$ = _____

e) $\sqrt{169}$ = _____ **f)** $\sqrt{196}$ = _____ **g)** $\sqrt{625}$ = _____ **h)** $\sqrt{1600}$ = _____

2. Write the largest square number that divides into each number.

a) 20 _____ **b)** 27 _____ **c)** 32 _____ **d)** 45 _____

e) 80 _____ **f)** 108 _____ **g)** 147 _____ **h)** 300 _____

B Simplifying a surd ●○○

Remember: $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$. Look for the largest square number that divides in.

3. Simplify.

a) $\sqrt{8}$ = _____ **b)** $\sqrt{28}$ = _____ **c)** $\sqrt{32}$ = _____ **d)** $\sqrt{63}$ = _____

e) $\sqrt{75}$ = _____ **f)** $\sqrt{44}$ = _____ **g)** $\sqrt{80}$ = _____ **h)** $\sqrt{108}$ = _____

i) $\sqrt{162}$ = _____ **j)** $\sqrt{300}$ = _____ **k)** $\sqrt{175}$ = _____ **l)** $\sqrt{242}$ = _____

4. Simplify.

a) $3\sqrt{20}$ = _____ **b)** $2\sqrt{45}$ = _____ **c)** $5\sqrt{12}$ = _____ **d)** $4\sqrt{18}$ = _____

C Adding and subtracting surds ●●○

Watch out: Only like surds can be added or subtracted, so simplify each surd first.

5. Simplify.

a) $2\sqrt{7} + 6\sqrt{7}$ **b)** $9\sqrt{3} - 4\sqrt{3}$ **c)** $5\sqrt{2} + \sqrt{2}$

d) $\sqrt{18} + \sqrt{2}$ **e)** $\sqrt{20} + \sqrt{45}$ **f)** $\sqrt{48} - \sqrt{12}$

g) $\sqrt{28} + \sqrt{63}$ **h)** $3\sqrt{2} + \sqrt{50}$ **i)** $\sqrt{98} - \sqrt{8}$

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D Multiplying and dividing surds ●●○

Remember: $\sqrt{a} \times \sqrt{b} = \sqrt{a \times b}$ and $\sqrt{a} \div \sqrt{b} = \sqrt{a \div b}$

6. Simplify.

a) $\sqrt{3} \times \sqrt{12}$ **b)** $\sqrt{2} \times \sqrt{7}$ **c)** $\sqrt{10} \times \sqrt{2}$

d) $3\sqrt{5} \times 2\sqrt{5}$ **e)** $2\sqrt{3} \times 5\sqrt{2}$ **f)** $\sqrt{11} \times \sqrt{11}$

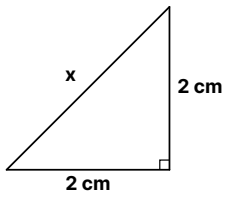
g) $\sqrt{45} \div \sqrt{5}$ **h)** $\sqrt{50} \div \sqrt{2}$ **i)** $8\sqrt{6} \div 4\sqrt{3}$

j) $2\sqrt{10} \times 3\sqrt{5}$ **k)** $\sqrt{98} \div \sqrt{2}$ **l)** $\sqrt{6} \times \sqrt{8}$

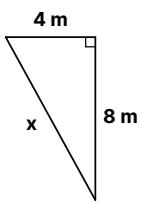
E Exact lengths ●●●

7. Find the length of the side marked x. Give each answer as a surd in simplest form. Diagrams are not to scale.

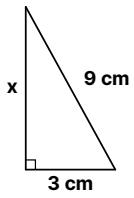
a)



b)



c)



F Mixed practice ●●●

8. Expand and simplify.

a) $\sqrt{3}(\sqrt{3} + 4)$ **b)** $\sqrt{2}(\sqrt{18} - 1)$ **c)** $(\sqrt{7} + 3)(\sqrt{7} - 3)$

d) $(2 + \sqrt{5})^2$ **e)** $3\sqrt{2}(\sqrt{2} + \sqrt{8})$ **f)** $(\sqrt{10} + \sqrt{3})(\sqrt{10} - \sqrt{3})$

9. A rectangle is $\sqrt{20}$ cm long and $\sqrt{5}$ cm wide. Give each answer in simplest form.

a) Find its area. **b)** Find its perimeter.

★ Extension

10. Rationalise the denominator: write each fraction without a surd on the bottom.

a) $\frac{1}{\sqrt{3}}$ **b)** $\frac{8}{\sqrt{2}}$ **c)** $\frac{14}{\sqrt{7}}$ **d)** $\frac{5}{2\sqrt{5}}$

11. A square has an area of 75 cm². Write the length of its sides in simplest surd form.

12. The two shorter sides of a right-angled triangle are $\sqrt{7}$ cm and $\sqrt{18}$ cm long. Find the length of the hypotenuse.

13. Show that $\sqrt{75} + \sqrt{12} = 7\sqrt{3}$.

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